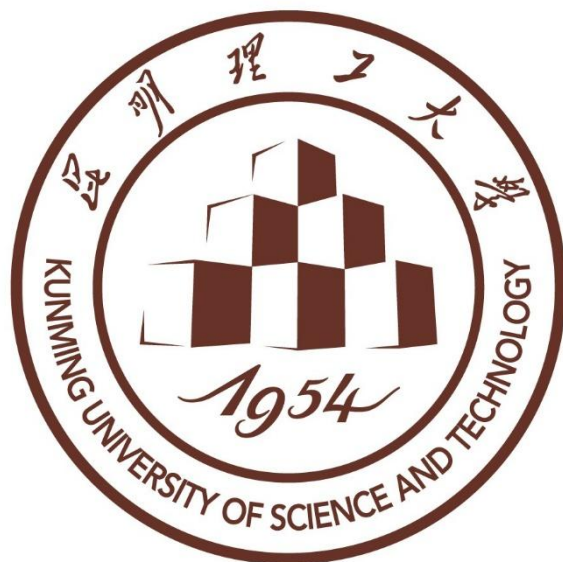


计算力学



程序设计作业

三维杆单元刚度矩阵与应力计算

姓 名 ： 孙宁泽

学 号 ： 202311012105

班 级 ： 工力 231

指导教师： 肖姚

1. 公式推导

1.1. 单元长度与方向余弦

$$\Delta x = x_2 - x_1, \quad \Delta y = y_2 - y_1, \quad \Delta z = z_2 - z_1$$

$$L = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$

$$l = \cos \alpha = \frac{\Delta x}{L}$$

$$m = \cos \beta = \frac{\Delta y}{L}$$

$$n = \cos \gamma = \frac{\Delta z}{L}$$

1.2. 单元轴向伸长量与节点位移之间的关系

$$\Delta L = (u_2 - u_1)c_x + (v_2 - v_1)c_y + (w_2 - w_1)c_z$$

1.3. 单元应变-位移矩阵

$$\varepsilon = \frac{1}{L} [-c_x \quad -c_y \quad -c_z \quad c_x \quad c_y \quad c_z] \mathbf{d}_e = \mathbf{B} \mathbf{d}_e$$

1.4. 单元刚度矩阵

$$\mathbf{k}_{\text{local}} = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\mathbf{T} = \begin{bmatrix} c_x & c_y & c_z & 0 & 0 & 0 \\ 0 & 0 & 0 & c_x & c_y & c_z \end{bmatrix}$$

$$\mathbf{K}_e = \mathbf{T}^T \mathbf{k}_{\text{local}} \mathbf{T}$$

$$\mathbf{K}_e = \frac{EA}{L} = \begin{bmatrix} c_x^2 & c_x c_y & c_x c_z & -c_x^2 & -c_x c_y & -c_x c_z \\ c_x c_y & c_y^2 & c_y c_z & -c_x c_y & -c_y^2 & -c_y c_z \\ c_x c_z & c_y c_z & c_z^2 & -c_x c_z & -c_y c_z & -c_z^2 \\ -c_x^2 & -c_x c_y & -c_x c_z & c_x^2 & c_x c_y & c_x c_z \\ -c_x c_y & -c_y^2 & -c_y c_z & c_x c_y & c_y^2 & c_y c_z \\ -c_x c_z & -c_y c_z & -c_z^2 & c_x c_z & c_y c_z & c_z^2 \end{bmatrix}$$

1.5. 单元应力和轴力计算格式

$$\varepsilon = \mathbf{B} \mathbf{d}_e, \quad \sigma = E \varepsilon, \quad N = \sigma A = EA \varepsilon$$

2. 程序实现

2.1. 单元刚度矩阵计算

```
# 单元刚度矩阵计算
# -----
3 usages
def truss3d_element_stiffness(x1, x2, E, A):

    # ===== 单元长度计算 =====
    dx = x2[0] - x1[0]
    dy = x2[1] - x1[1]
    dz = x2[2] - x1[2]
    L = np.sqrt(dx**2 + dy**2 + dz**2)

    # ===== 方向余弦 =====

    cx, cy, cz = dx/L, dy/L, dz/L

    T = np.array([[cx, cy, cz, 0, 0, 0],
                  [0, 0, 0, cx, cy, cz]])

    k_local = (E * A / L) * np.array([[1, -1], [-1, 1]])
    Ke = T.T @ k_local @ T

    return L, (cx, cy, cz), Ke
```

2.2. 应变、应力与轴力计算

```
# ===== 应变/应力/轴力计算 =====  
2 usages  
def truss3d_element_stress(x1, x2, E, A, de):  
  
    L, (cx, cy, cz), _ = truss3d_element_stiffness(x1, x2, E, A)  
    B = np.array([-cx, -cy, -cz, cx, cy, cz]) / L  
  
    epsilon = B @ de  
    sigma = E * epsilon  
    N = sigma * A  
    return epsilon, sigma, N
```

3. 程序验证

3.1. 算例一：

算例1：沿x轴一维杆单元

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单元长度 $L = 2.0000$ m

方向余弦 $(cx, cy, cz) = (1.0, 0.0, 0.0)$

刚度矩阵 K_e (6x6):

[[10000000.	0.	0.	-10000000.	0.	0.]
[	0.	0.	0.	0.	0.]
[	0.	0.	0.	0.	0.]
[-10000000.	0.	0.	10000000.	0.	0.]
[	0.	0.	0.	0.	0.]
[	0.	0.	0.	0.	0.]]

应变 $\epsilon = 0.000500$

应力 $\sigma = 100.00$ MPa

轴力 $N = 10000.00$ N

3.2. 算例二:

算例2: 空间任意方向杆单元

=====

单元长度 $L = 3.0000 \text{ m}$

方向余弦 $(cx, cy, cz) = (0.3333333333333333, 0.6666666666666666, 0.6666666666666666)$

刚度矩阵 $K_e (6 \times 6)$:

```
[ [ 1555555.5556  3111111.1111  3111111.1111 -1555555.5556 -3111111.1111
   -3111111.1111]
 [ 3111111.1111  6222222.2222  6222222.2222 -3111111.1111 -6222222.2222
   -6222222.2222]
 [ 3111111.1111  6222222.2222  6222222.2222 -3111111.1111 -6222222.2222
   -6222222.2222]
 [-1555555.5556 -3111111.1111 -3111111.1111  1555555.5556  3111111.1111
   3111111.1111]
 [-3111111.1111 -6222222.2222 -6222222.2222  3111111.1111  6222222.2222
   6222222.2222]
 [-3111111.1111 -6222222.2222 -6222222.2222  3111111.1111  6222222.2222
   6222222.2222]]
```

应变 $\varepsilon = 0.001000$

应力 $\sigma = 210.00 \text{ MPa}$

轴力 $N = 42000.00 \text{ N}$

4. 刚度矩阵物理意义验证

任务4: 刚度矩阵物理意义验证 (第4列)

=====

节点力 $F_e = [-1555555.5556 -3111111.1111 -3111111.1111 1555555.5556 3111111.1111$
 $3111111.1111]$

结论: F_e 等于 K_e 的第4列, k_{ij} 表示j自由度单位位移引起的i自由度力

